

General question.

①

What is the shape of the discreteness locus

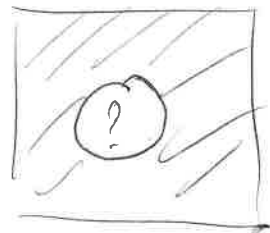
in $\text{Hom}(\Gamma, \text{PSL}(2, \mathbb{C}))$ containing a fixed ^{discrete} Γ ?

- Remark. The analogous qn for $\mathbb{R}$ is "boring" since the discreteness locus is clopen, i.e. on entire connected component.

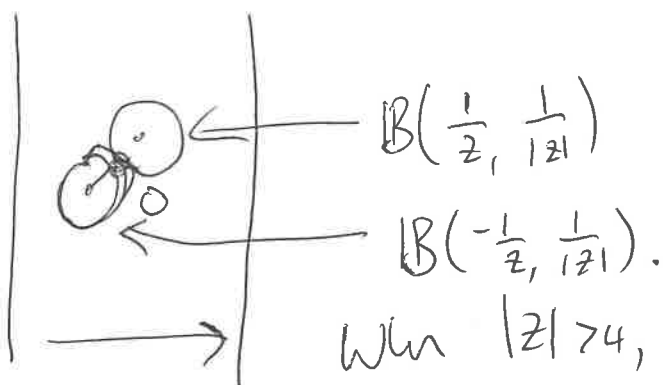
Part 1. Ping-pong results.

Brenner (1955): When $|z| > 4$,

$G_z = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle$
is discrete and free.



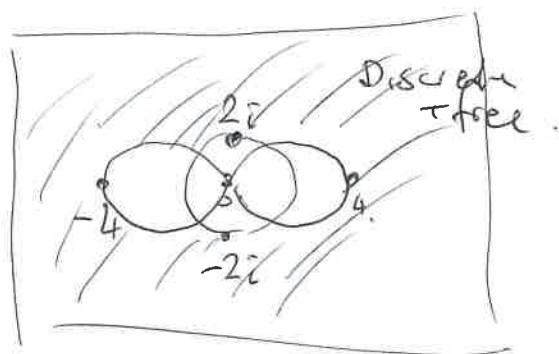
Proof. Here is a f.d. for $G_z \curvearrowright \mathbb{C}$:



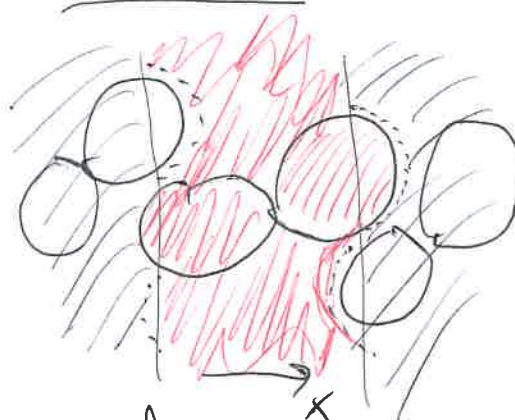
When $|z| > 4$, these balls lie inside the vertical strip of width 1.

Chang - Jennings - Ree (1958):

②



Proof:

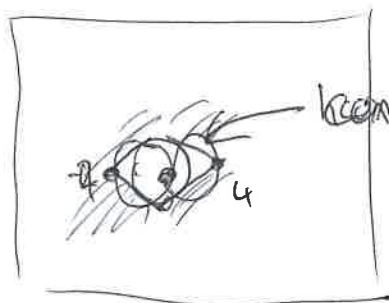


Need (i) X to send $\blacksquare$ into $\square$
 (ii) Y to send $\square$ into $\blacksquare$.

Each + case $\text{Isom}^{\pm}(Y)$ are disjoint from the X_k^{th} translates.

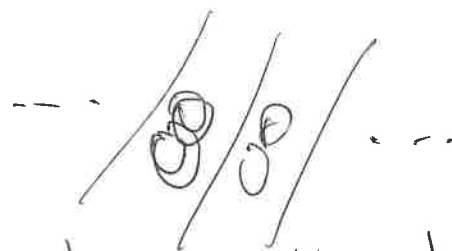
Rmk. Pigeon-hole is needed: - we have an explicit f. domain.

Lyndon - Ullmann (1999):



$\text{ker } \nu. (B(0,1) \cup \{\pm 4\})$.

Proof: modify pigeon-hole
 set to let the vertical
 strip be "slanted" to
 match the angle of
 discs.



(ie. change z-variable until we are in the vertical-strip setting.)

E - Gong - Martin - Schiller (24):

(3)

- generalize to elliptic gens of order p, q .
- blindly following Chac - Sen - Ree, Lyda - Ullmann giving ~~that~~ suboptimal result if $p \neq q$, so more work is done to sort it out.

Thm. $\exists$ a dodecagon $\Omega_{p,q}$ in $\mathbb{C}$ such that
if $z \in \Omega_{p,q}$, then

$$\left\langle \begin{pmatrix} e^{i\pi/p} & 1 \\ 0 & e^{-i\pi/p} \end{pmatrix}, \begin{pmatrix} e^{i\pi/q} & 0 \\ z & e^{-i\pi/q} \end{pmatrix} \right\rangle$$

$$\simeq \mathbb{Z}/p\mathbb{Z} \# \mathbb{Z}/q\mathbb{Z} \text{ is dense.}$$

Application 1. ~~if it is known that all arithmetic~~

Enumerating all arithmetic subgroups of $\mathrm{PSL}(2, \mathbb{C})$ on two elliptic generators. If bounds on degree of trace field are known, then there are only finitely many candidates in any computer set. $\Omega_{p,q}$ provides such a set.

Application 2. A conjecture of Morier-Genoud, ~~Ovsienko~~ Ovsienko, Veselov (IMRN '24):

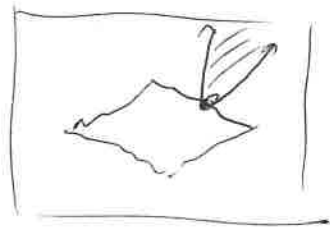
the specialized Buan rep $B_3 \rightarrow \langle \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ t & t \end{pmatrix} \rangle$
is faithful for t outside the annulus $\frac{3-\sqrt{5}}{2} \leq |t| \leq \frac{3+\sqrt{5}}{2}$.

(applicable to q -rational numbers).

These bounds are basically nbds of ∞ . They do not approximate the boundary of the deformation well. & it is hard to refine them. (4)

2. The structure theory of $\mathcal{R}$.

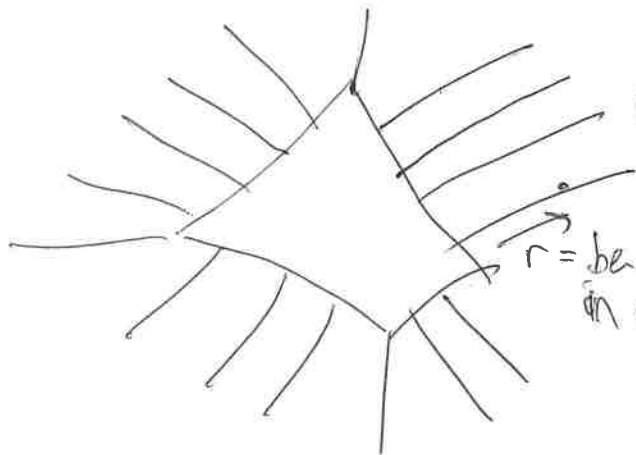
Goal: Find neighborhoods of boundary points.



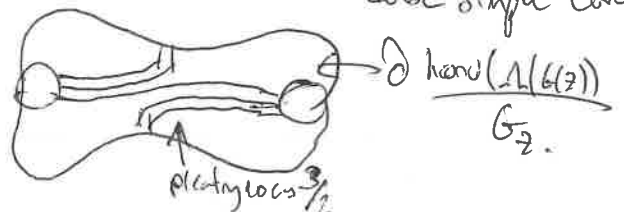
Need geometric motivation for what a "good" choice would be: fundamental domains given by Bom. Circles like in (1) are arbitrarily complicated.

Theorem (Keen-Series, 1994 following R. Riley, c.1979)

(i) There is a natural ^{radial} coordinate system on $\mathcal{R}$ (the discrete-fathful ~~subset~~ subset of $\langle (0,1), (\frac{1}{2},1) \rangle$):



θ = choice of maximal lamination on $S_{0,4}$ (i.e. simple closed curve or dense simple curve)
 r = bending angle in curves core.



$\partial \text{hom}(\Lambda(6,2))$
 G_2 .

pleating locus $\frac{3}{2}$

(ii) each radial leaf (pleaty ray) coming from a rational lamination ν is of a con. compact of a semi-aly. set

the form $f_{p/q}^{-1}([-2, -\infty))$

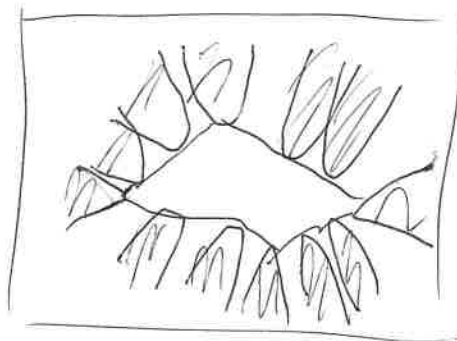
where f is a polynomial in z (actually, f is the trace of the p/q slope word).

This motivates:

Theorem (E. - Martin-Schulz, '28): ~~the same~~
The "same" connected component of

$$f_{p/q}^{-1}(\{z \in \mathbb{C} : \operatorname{Re} z < \frac{1}{2}\})$$

is a neighbourhood of the p/q cusp point on ∂R .



i.e. $\exists$ a cbl collection of semi-aly. subsets that exhaust R .
 $\nexists$

Corollary: there exists a semi-algorithm for membership in R .

Actually, this is a generalisation.

(6)

Thm (E., '25). Let G be a geom. finite Kleinian group, with $\Omega(G) \neq \emptyset$. Then the q.c. deformation space

$$QH(G) = \left\{ \rho \in \text{Hom}(G, \text{PSL}(2, \mathbb{C})) : \begin{array}{l} \rho \text{ is discrete, faithful,} \\ \text{type preserving, and} \\ \rho(G) \text{ q.c. -conjugate to } G \end{array} \right\}$$

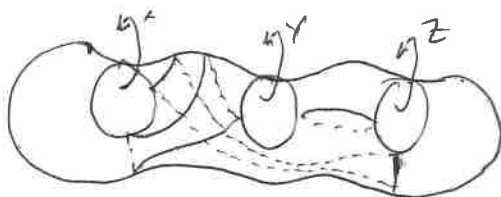
decomposes as a union of atably many semialgebraic sets in $\text{Hom}(G, \text{PSL}(2, \mathbb{C}))$.

I will sketch the construction ~~for~~ for a special case: $\Omega(G)/G$ is an n -punctured sphere, & there are no other ends.

STEP 1. Fix a basepoint $G_0 \in QH(G)$ s.t. the pleating locus of ∂ leaves $\Omega(G_0)/G_0$ is a rational lamination (= cusp support). Then $\Omega(G_0)$ ~~is~~ cut out a pattern of overlapping discs.

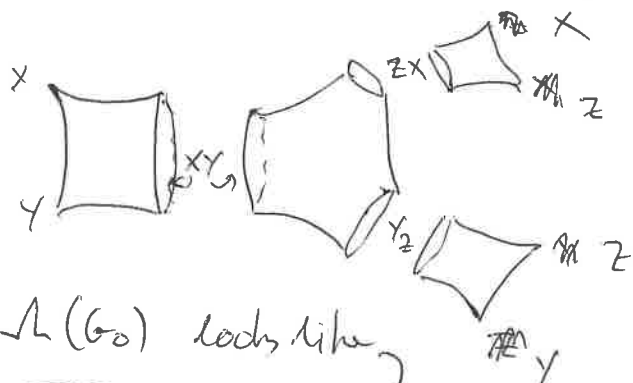
Eg.

(7)

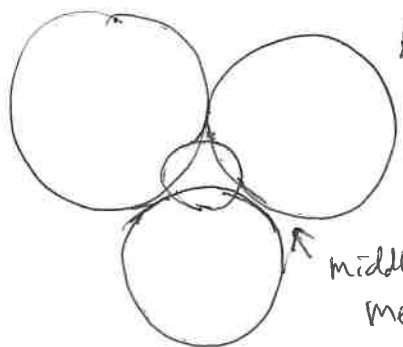


Σ is represented by x, y , and z .

The decomposition into flat pieces is:



Then $\Sigma(G_0)$ looks like

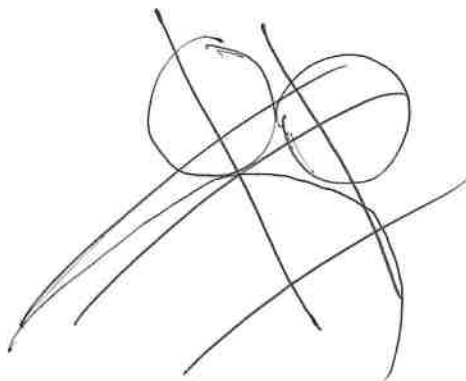
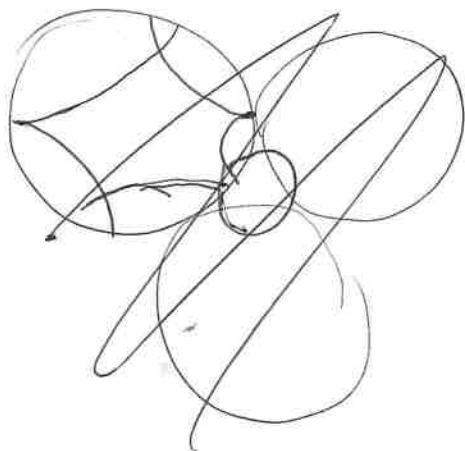


middle circle does NOT meet tangent points.

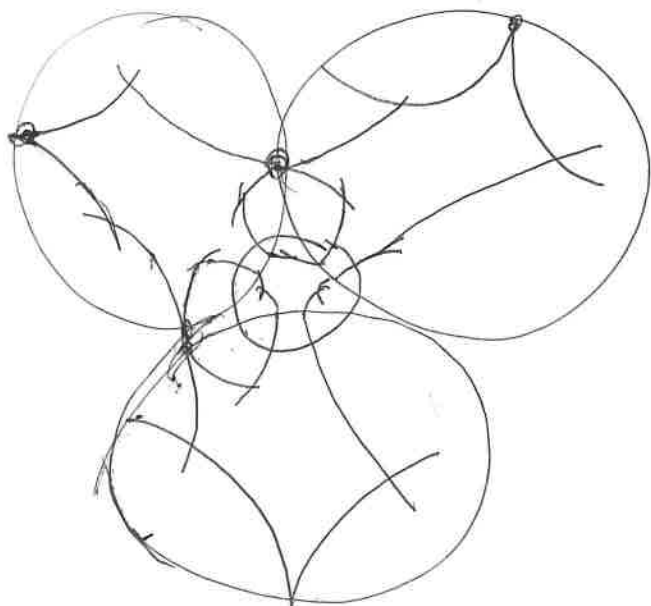
Put together with translates by other simple closed curves transverse to those of Σ .

We refer to such a "fundamental domain" for Σ as a "circle chain" (cf. Keen-Saris).

Step 2. Draw the desired fundamental domain for each.



STEP 2. Draw the obvious fundamental domain (8) for each flat piece (given by isometric circles for each ∂ -parallel element).



(Here, if we had a surface with genus, we run into trouble with the "not matching up around cycles" $\frac{1}{2}$ Here, we have a tree



STEP 4. Show that this domain can be made to vary algebraically ~~within~~ (e.g. on traces) for all groups Γ within pleating locus $\mathcal{L}$. (= same pattern of circles).

Here, observe that circles in $\mathcal{L}$ are parameterized by ~~the~~ point in $\mathbb{P}^3$, & pencils (e.g. spanned by non-circs) are embedded $\mathbb{P}^1$'s in $\mathbb{P}^3$.

Step 4. We have an algebraic

(4)

$$\text{map } \phi_\Lambda : X(G, \text{PSL}(2, \mathbb{A})) \rightarrow (\mathbb{P}^3)^N$$

which, on the subset consisting of groups with plenty loci Λ , assigns to every group a fundamental domain.

~~Since it is~~

(We know it is algebraic on the subset, but it may not extend).

Write down conditions $*$ in trace parameters $\rightarrow$ (using Ralg. geom. of circles)

such that the image of ϕ_Λ is a fundamental domain. These give a full dimensional semi-algebraic ~~sets~~ set in which $\exists$ a fundamental domain for $\tilde{G} \in X(G, \text{PSL}(2, \mathbb{A}))$ with the same combinatorics.

Since the ~~set~~ of groups with plenty loci rational laminations is dense ~~in~~ in $\mathcal{QH}(G)$, we have covered $\mathcal{QH}(G)$ with semi-aly. sets, one for each lamination λ . $\square$.

Remark. - Some technicalities involved to go from $(\mathbb{P}^3)^N$ to "actual" fundamental domains (need to keep track of twists of circles with parameters).
- Actually constructing the semi-aly. sets is technically possible (fully constructive) but infeasible for "interesting" exs.)

Qns.

— Can we extend this to the full $1H^3$ -fwd. dom?

Stably embedded plethl subgroups
 "lower & lower" families of circles in $\mathcal{L}(G)$;
 need to justify a hierarchy of these
 nested circle paths.

— Lot of flex & choice involved — can
 the constraints be made more quantitative
 Canonical?